



Probability Generating Functions · Worksheet 1

Further Statistics 1 Chapter 7: Probability Generating Functions. Answer all questions. Show your working clearly. Total: 67 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 7. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- $G_X(t) = E(t^X)$. Use $E(X) = G'_X(1)$ and $\text{Var}(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2$. Give exact answers where possible.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Probability generating functions	20	
B Mean and variance from a PGF	21	
C Sums of independent random variables and uniqueness	21	
Roar-level challenge	5	
Total	67	

Section A Probability generating functions

20 marks

1. The discrete random variable X has the probability distribution below.

[2]

x	1	2	3
$P(X = x)$	0.1	0.6	0.3

Write down the probability generating function of X .

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2. The probability generating function of X is $G_X(t) = k(2+t)^2$, where k is a constant. Find the value of k and $P(X = 1)$.

[2]

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3. The random variable $X \sim \text{Po}(\lambda)$. Show that the probability generating function of X is $G_X(t) = e^{\lambda(t-1)}$. [3]

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4. The random variable X has a geometric distribution with $P(X = x) = q^{x-1}p$ for $x = 1, 2, 3, \dots$, where $q = 1 - p$. [4]

Show that $G_X(t) = \frac{pt}{1 - qt}$, and state the set of values of t for which this is valid.

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5. The probability generating function of the discrete random variable X is

$$G_X(t) = \frac{t^2}{3 - 2t}.$$

- (a) Find $P(X = 2)$ and $P(X = 3)$. [2]

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- (b) By writing $G_X(t)$ as a series in powers of t , find an expression for $P(X = x)$, stating the possible values of x . [3]

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- (c) State the set of values of t for which $G_X(t)$ is valid. [2]

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- (d) By considering $G_X(1)$ and $G_X(-1)$, find the probability that X is even. [2]

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(Total for Question 5 is 9 marks)



Section B Mean and variance from a PGF

21 marks

6. The probability generating function of X is $G_X(t) = 0.2 + 0.3t + 0.5t^2$. [2]

Use $G_X(t)$ to find $E(X)$.

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7. The random variable $X \sim B(n, p)$ has probability generating function $G_X(t) = (q + pt)^n$, where $q = 1 - p$. [3]

Use $G_X(t)$ to show that $E(X) = np$.

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8. The probability generating function of X is $G_X(t) = \frac{1}{27}(1 + 2t)^3$. [3]

Find $E(X)$ and $\text{Var}(X)$.

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9. The random variable $X \sim \text{Po}(\lambda)$, with $G_X(t) = e^{\lambda(t-1)}$. [4]

Use $G_X(t)$ to show that $E(X) = \lambda$ and $\text{Var}(X) = \lambda$.

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10. The probability generating function of the discrete random variable X is

$$G_X(t) = \frac{t + t^2}{4 - 2t}.$$

(a) Find $P(X = 1)$ and $P(X = 2)$. [2]

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(b) Show that $E(X) = \frac{5}{2}$. [3]

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(c) Find $\text{Var}(X)$. [4]

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(Total for Question 10 is 9 marks)

**Section C Sums of independent random variables and uniqueness**

21 marks

11. The independent random variables X and Y are such that $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(3)$. [2]

Use probability generating functions to show that $X + Y \sim \text{Po}(5)$.

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12. The random variable X has probability generating function $G_X(t) = 0.3 + 0.7t$. The random variable Y is the sum of n independent observations of X . [3]

Find the probability generating function of Y , and hence identify the distribution of Y .

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13. The random variable $X \sim \text{Po}(4)$, and $Y = 3X + 2$. [3]

Write down the probability generating function of Y , and use it to find $E(Y)$.

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14. The independent random variables X and Y have probability generating functions [4]

$$G_X(t) = \frac{1}{3}(1 + 2t) \quad \text{and} \quad G_Y(t) = \frac{1}{3}(1 + t + t^2).$$

Find the probability generating function of $X + Y$, and hence find $P(X + Y = 2)$ and $E(X + Y)$.

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15. A player throws darts at a target until she hits it. The number of throws needed, X , has a geometric distribution with parameter 0.4, so that

$$G_X(t) = \frac{0.4t}{1 - 0.6t}.$$

She does this three times, independently. The random variable Y is the total number of throws needed to hit the target three times.

- (a) Find the probability generating function of Y . [2]

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- (b) State, with a reason, the distribution of Y , including the values of any parameters. [2]

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- (c) Use the probability generating function of Y to find $E(Y)$. [3]

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- (d) Find $P(Y = 4)$. [2]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. The score, X , on a fair six-sided die has probability generating function $G_X(t) = \frac{t(1-t^6)}{6(1-t)}$. [5]
The die is rolled three times. Use probability generating functions to find the probability that the total score is 10.

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END OF WORKSHEET

Total: 67 marks